LECTURE 16 · DISCRETE MATHEMATICS

Order without a single ranking

Partial orders, Hasse diagrams, and extremal elements

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A guided set of course notes

An order need not place every pair in a race. These notes explain how divisibility and set containment create legitimate orders with incomparable elements, how Hasse diagrams compress them, and why “maximal” is not the same as “greatest.”

How to read these notes. Slide-derived material is presented as the main narrative. Any new context, extension, or extra example is labeled **[Added]**. Any unresolved gap in the slide material is labeled **[Unclear]**.

Based on Lecture 16 and Rosen, 7th ed., §§9.6 and 2.1–2.2

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Learning goals

After working through these notes, you should be able to:

- prove or disprove that a relation is a partial order;
- distinguish comparability, incomparability, partial order, and total order;
- construct a Hasse diagram from a finite partial order and recover the relation from the diagram;
- identify all minimal and maximal elements and decide whether least or greatest elements exist;
- explain why a greatest or least element, if present, is unique; and
- use complement pairing to solve the optional intersecting-family challenge.

The central change from Lecture 15

An equivalence relation uses symmetry to say that two objects belong to the same class. A partial order uses antisymmetry instead: two distinct objects cannot point to each other. The result is not necessarily a single ranking. Some elements can remain genuinely incomparable.

1 Warm-up: an averaging proof on a circle

Slide sequence: Slides 2–3

Arrange the numbers $1, 2, \ldots, 10$ around a circle. Must some three consecutive entries have sum greater than 17? The smallest possible triple sum is $1 + 2 + 3 = 6$, and the largest is $8 + 9 + 10 = 27$, but these observations alone do not prove anything about *every* arrangement.

The useful move is to cut the circle immediately after the entry 1. Read the remaining entries in circular order as $b_1, b_2, \ldots, b_9$. The three blocks

$$(b_1, b_2, b_3), \qquad (b_4, b_5, b_6), \qquad (b_7, b_8, b_9)$$

are all triples of consecutive entries on the original circle. Their sums T_1, T_2, T_3 satisfy

$$T_1 + T_2 + T_3 = 2 + 3 + \cdots + 10 = 54.$$

If all three sums were at most 17, their total would be at most 51, a contradiction. Equivalently, at least one triple reaches the average:

$$\max(T_1, T_2, T_3) \geq \frac{54}{3} = 18 > 17.$$

Takeaway. A global total can force a large local block. The proof succeeds because the cut at 1 partitions the other nine numbers into three valid consecutive triples.

[Added] Worked example: the averaging bound for twelve numbers

Arrange $1, 2, \ldots, 12$ around a circle. What lower bound follows immediately for the largest sum of three consecutive entries?
Group the circle into four disjoint consecutive triples. Their total is $1+2+\cdots+12 = 78$, so one triple has sum at least

$$\left\lceil \frac{78}{4} \right\rceil = 20,$$

because triple sums are integers. The calculation teaches an important proof habit: compute exactly what the chosen partition and its invariant guarantee.

2 Partial orders: structure without universal comparison

Slide sequence: Slides 4–11

Textbook anchor: §9.6, Definitions 1–3 and Examples 1–7, pp. 618–620.

A relation $\preceq$ on a set S is a **partial order** when it is:

1. **reflexive:** $a \preceq a$ for every $a \in S$;
2. **antisymmetric:** $a \preceq b$ and $b \preceq a$ imply $a = b$; and
3. **transitive:** $a \preceq b$ and $b \preceq c$ imply $a \preceq c$.

A set equipped with a partial order is a **partially ordered set**, or **poset**, written $(S, \preceq)$. The symbol $\preceq$ is generic: it may mean numerical comparison, divisibility, containment, a prerequisite relation, or something else. Always read its definition before interpreting "above" or "below."

2.1 Three model posets

2.1.1 Reverse numerical order

The usual relation $\geq$ is a partial order on $\mathbb{Z}$: $a \geq a$; the two inequalities $a \geq b$ and $b \geq a$ force equality; and $a \geq b \geq c$ gives $a \geq c$. Reversing the direction of a familiar order does not destroy its order properties.

2.1.2 Divisibility

On the positive integers, define $a \preceq b$ when $a \mid b$.

- **Reflexive:** $a = a \cdot 1$, so $a \mid a$.
- **Antisymmetric:** if $b = ak$ and $a = b\ell$ for positive integers k, ℓ, then $a = a(k\ell)$. Since $a > 0$, $k\ell = 1$, so $k = \ell = 1$ and $a = b$.
- **Transitive:** if $b = ak$ and $c = b\ell$, then $c = a(k\ell)$, so $a \mid c$.

The restriction to positive integers matters. On all of $\mathbb{Z}$, $2 \mid -2$ and $-2 \mid 2$, although $2 \neq -2$; antisymmetry fails.

2.1.3 Set containment

On $\mathcal{P}(U)$, define $A \preceq B$ when $A \subseteq B$. Every set is contained in itself; mutual containment implies equality; and containment chains. Therefore $(\mathcal{P}(U), \subseteq)$ is a poset.

[Added] Worked example: task prerequisites form a partial order

Let S be the tasks in a project. Define $x \preceq y$ when $x = y$ or task x must be completed before task y can begin. Reflexivity comes from the explicit equality case. Antisymmetry expresses that two distinct tasks cannot each be prerequisites of the other in a feasible plan. Transitivity says that if x precedes y and y precedes z, then x precedes z. Two tasks in independent branches may be incomparable. This application is motivated by Rosen §9.6, p. 618.

2.2 Efficiently disproving a partial order

To prove that a relation is a partial order, verify all three properties. To disprove it, one failed property and one clear counterexample are enough.

Relation and set	Failed property	Witness
$<$ **on** $\mathbb{Z}$	Reflexivity	$a < a$ is false for every a.
Divisibility on $\mathbb{Z}$	Antisymmetry	$2 \mid -2$ and $-2 \mid 2$, but $2 \neq -2$.
Congruence modulo 3 **on** $\mathbb{Z}$	Antisymmetry	$1 \equiv 4 \pmod 3$ in both directions, but $1 \neq 4$.

[Added] Worked example: "shares a factor" is not transitive

On integers greater than 1, define aRb when $\gcd(a, b) > 1$. The relation is reflexive and symmetric. It is not transitive:

$$\gcd(6, 10) = 2, \qquad \gcd(10, 15) = 5, \qquad \gcd(6, 15) = 3 > 1.$$

This first choice accidentally remains related. A useful correction is $a = 6, b = 10, c = 25$:

$$\gcd(6, 10) = 2, \quad \gcd(10, 25) = 5, \quad \gcd(6, 25) = 1.$$

Thus $6R10$ and $10R25$, but $6 \not R 25$. The corrected witness illustrates why counterexamples must be checked, not merely guessed.

2.3 Comparable, incomparable, and totally ordered

Elements a, b of a poset are **comparable** when $a \preceq b$ or $b \preceq a$. Otherwise they are **incomparable**. In the divisibility poset:

- 3 and 9 are comparable because $3 \mid 9$;
- 5 and 7 are incomparable; and
- 8 and 12 are incomparable even though $8 < 12$ numerically.

A **total order** or **linear order** is a partial order in which every pair is comparable. A totally ordered subset is also called a **chain**. The usual $\leq$ on $\mathbb{Z}$ is total. Divisibility on $\mathbb{Z}^+$ is partial but not total.

Exam lens: "partial" does not mean "missing a property"

A partial order satisfies all three required properties. The word "partial" means only that some distinct elements may be incomparable. A total order is a special kind of partial order, not a competing definition.

3 Hasse diagrams: keep only essential comparisons

Slide sequence: Slides 12–19

Textbook anchor: §9.6, "Hasse Diagrams," pp. 622–624.

For a poset, write $x \prec y$ when $x \preceq y$ and $x \neq y$. We say that y **covers** x when

$$x \prec y \quad \text{and there is no } z \text{ with } x \prec z \prec y.$$

The **covering relation** contains exactly these essential adjacent comparisons.

A **Hasse diagram** for a finite poset is produced by:

1. deleting all loops, because reflexivity is understood;
2. deleting transitive shortcuts, keeping only covers;
3. placing the greater element above the lesser element; and
4. removing arrowheads, because every edge is read upward.

In a Hasse diagram, $x \preceq y$ exactly when $x = y$ or an upward path leads from x to y. Physical height alone is not enough: there must be an upward path.

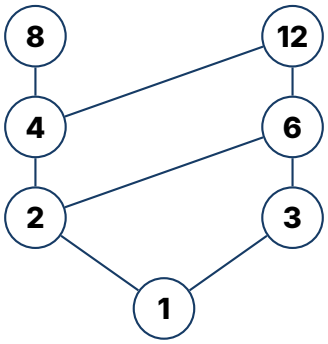
3.1 The lecture's divisibility diagram

Let

$$S = \{1, 2, 3, 4, 6, 8, 12\}$$

ordered by divisibility. The covering pairs are

$$(1,2), (1,3), (2,4), (2,6), (3,6), (4,8), (4,12), (6,12).$$



The diagram has two branches. The path $2 \to 4 \to 12$ shows $2 \mid 12$ even though no direct edge is drawn. By contrast, no upward path connects 3 and 4, so they are incomparable. The vertex 8 is drawn at the same height as 12, but that visual choice does not assert a comparison.

The transitive pairs deleted from the original directed graph include

$$(1,4), (1,6), (1,8), (1,12), (2,8), (2,12), (3,12).$$

For example, $(1,4)$ is redundant because of the path $1 \to 2 \to 4$.

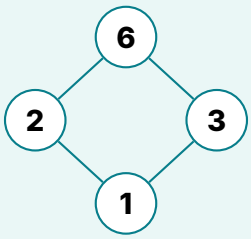
Worked example: build a diamond from scratch

For $S = \{1, 2, 3, 6\}$ under divisibility, begin by listing all strict comparisons:

$$1 \mid 2, quad1 \mid 3, quad1 \mid 6, quad2 \mid 6, quad3 \mid 6.$$

The pair $(1,6)$ is not a cover because $1 \mid 2 \mid 6$ (and also $1 \mid 3 \mid 6$). The covers are therefore

$$(1,2), (1,3), (2,6), (3,6).$$



The diagram is a diamond. Its four edges contain enough information to recover the missing comparison $1 \mid 6$ by transitivity.

3.2 Recovering the original relation

Let C be the upward covering relation of a finite poset on S. Restore the partial order in two steps:

1. take the transitive closure of C, adding a pair for every positive-length upward path;
2. add the diagonal Δ_S to restore reflexivity.

In relation notation,

$$R = \Delta_S \cup C \cup C^2 \cup C^3 \cup \cdots.$$

Because S is finite, only finitely many path lengths are needed.

[Added] Worked example: the subset lattice on two elements

Order $\mathcal{P}(\{a, b\})$ by containment. The covers are

$$\varnothing \prec \{a\}, \quad \varnothing \prec \{b\}, \quad \{a\} \prec \{a, b\}, \quad \{b\} \prec \{a, b\}.$$

This is another diamond. The omitted relation $\varnothing \subseteq \{a, b\}$ is recovered by either upward path. For three underlying elements, the Hasse diagram has levels by set size: the empty set, three singletons, three two-element sets, and the full set. This elaborates Rosen §9.6, Example 13 and Figure 4, pp. 623–624.

Exam lens: a safe Hasse-diagram workflow

List all strict comparison pairs first. For each (x, y), ask whether some z lies strictly between them. Draw an edge only when the answer is no. This prevents two common errors: drawing reflexive loops and retaining transitive shortcuts.

4 Extrema: local tops versus global top

Slide sequence: Slides 20–28 and 31

Textbook anchor: §9.6, "Maximal and Minimal Elements," pp. 624–625.

Let $(S, \preceq)$ be a poset.

- An element x is **minimal** if no distinct y satisfies $y \preceq x$.
- An element x is **maximal** if no distinct y satisfies $x \preceq y$.
- An element ℓ is the **least element** if $\ell \preceq y$ for every $y \in S$.
- An element g is the **greatest element** if $y \preceq g$ for every $y \in S$.

Minimal and maximal are local conditions: nothing lies strictly below or above the element. Least and greatest are global: the element must compare with everyone. A least element is minimal and a greatest element is maximal, but the converses can fail.

4.1 The divisibility example resolves the opening question

In

$$(\{1, 2, 3, 4, 6, 8, 12\}, |),$$

the only minimal element is 1. It is also the least element because it divides every member of the set. The maximal elements are 8 and 12. Neither divides the other, so neither is greatest. There is no greatest element.

This answers the cover question: yes, 8 and 12 can both sit at the top. They are maximal relative to divisibility even though $8 < 12$ numerically. The integer 24 would be an upper bound in the larger poset of positive integers, but it cannot be greatest here because $24 \notin S$.

Term	Test	Possible multiplicity
Minimal	Nothing distinct lies below	Several may exist
Maximal	Nothing distinct lies above	Several may exist
Least	Lies below every element	At most one
Greatest	Lies above every element	At most one

4.2 Why least and greatest elements are unique

Suppose g and h are both greatest. Since h is greatest, $g \preceq h$. Since g is greatest, $h \preceq g$. Antisymmetry gives $g = h$. Reversing all comparisons proves uniqueness of a least element.

The proof does not apply to maximal elements. Two maximal elements need not be comparable, so we cannot obtain both directions and invoke antisymmetry.

4.3 Four possible global outcomes

The lecture shows four Hasse diagrams illustrating these possibilities:

1. a least element a but no greatest element;
2. neither a least nor a greatest element;
3. a greatest element d but no least element; and
4. both least element a and greatest element d.

[Unclear] one interior vertex in the first source diagram is unlabeled

The slide source notes that one interior vertex in the first legacy diagram has no visible name. Its name is not needed to determine the extrema: the bottom vertex a reaches every element upward, while multiple incomparable top vertices prevent a greatest element. The missing label affects neither answer, but a spoken explanation may have supplied the intended name.

4.4 The underlying set changes the answer

In $(\mathcal{P}(U), \subseteq)$, the empty set is least and U is greatest. In $(\mathbb{Z}^+, |)$, 1 is least but there is no greatest element: every n lies below the distinct element $2n$.

Infinite orders may have no extrema at all. In $(\mathbb{Z}, \leq)$, every n has $n-1$ below and $n+1$ above, so no element is minimal or maximal. In $(\mathbb{Z}^+, |)$, there is no maximal element because $n \mid 2n$ and $n \neq 2n$ for every n.

[Added] Worked example: several minimal and maximal elements

Consider

$$S = \{2, 4, 5, 10, 12, 20, 25\}$$

under divisibility. The minimal elements are 2 and 5: neither has a smaller divisor belonging to S. The maximal elements are $12, 20, 25$: none divides a distinct member of S. There is no least element because neither 2 nor 5 divides the other, and no greatest element because the three maximal elements are incomparable. This example is adapted from Rosen §9.6, Example 14, p. 624.

[Added] Worked example: greatest depends on membership, not imagination

In $S = \{2, 3, 6\}$ under divisibility, 6 is greatest. In $T = \{2, 3, 6, 10\}$, 6 remains maximal, but it is no longer greatest because 10 is incomparable with 6. The order relation did not change; enlarging the underlying set changed the global answer.

5 Optional challenge: intersecting families of subsets

Slide sequence: Slides 29–30

Textbook anchor: §2.1, pp. 121–122; §2.2, p. 129.

Let $U = \{1, \dots, n\}$ with $n \geq 1$, and let $A_1, \dots, A_k$ be distinct subsets such that

$$A_i \cap A_j \neq \varnothing \qquad \text{for every } i, j.$$

Such a collection is an **intersecting family**. We will show

$$k \leq 2^{n-1}.$$

Pair every subset $A \subseteq U$ with its complement $U \setminus A$. A subset never equals its own complement when U is nonempty. The 2^n subsets therefore form 2^{n-1} disjoint complement pairs.

An intersecting family can contain at most one member of each pair, because

$$A \cap (U \setminus A) = \varnothing.$$

Hence $k \leq 2^{n-1}$.

The bound is attainable: choose every subset containing the element 1. There are 2^{n-1} such subsets, because each of the other $n - 1$ elements may be included or excluded independently. Any two selected subsets intersect at 1.

Worked example: see the complement pairs when $n = 3$

Let $U = \{1, 2, 3\}$. The eight subsets form four complement pairs:

$$\varnothing \leftrightarrow \{1, 2, 3\}, \qquad \{1\} \leftrightarrow \{2, 3\},$$
$$\{2\} \leftrightarrow \{1, 3\}, \qquad \{3\} \leftrightarrow \{1, 2\}.$$

The family of all subsets containing 1 is

$$\{\{1\}, \{1, 2\}, \{1, 3\}, \{1, 2, 3\}\},$$

which selects exactly one subset from every complement pair and has the maximum size $2^{3-1} = 4$.

6 Summary

1. A partial order is reflexive, antisymmetric, and transitive. It models direction without requiring every pair to be comparable.
2. Divisibility on positive integers and containment on a power set are central examples. The underlying set is part of the definition and can change whether antisymmetry or extrema exist.
3. Comparable elements are related in at least one direction. Incomparable elements are neither equal nor misordered; the relation simply supplies no direction between them.
4. A total order is a partial order in which every pair is comparable.
5. A Hasse diagram removes loops and transitive shortcuts. An upward path, not mere height, represents a comparison.

6. A cover $x \prec y$ has no element strictly between its endpoints. The covers determine a finite poset through reflexive transitive closure.
7. Minimal and maximal are local conditions. Least and greatest compare globally with every element and, when they exist, are unique.
8. A finite poset may have several minimal or maximal elements but at most one least or greatest element.
9. Complement pairing proves that an intersecting family of subsets of an n-element set has at most 2^{n-1} members.

7 Key terms glossary

Antisymmetric	$a \preceq b$ and $b \preceq a$ imply $a = b$.
Chain	A totally ordered subset; every two elements in it are comparable.
Comparable	Two elements a, b for which $a \preceq b$ or $b \preceq a$.
Cover	y covers x when $x \prec y$ and no element lies strictly between them.
Covering relation	The set of all cover pairs in a poset.
Greatest element	An element g satisfying $x \preceq g$ for every element x.
Hasse diagram	A finite-poset diagram showing only cover edges, read upward.
Incomparable	Two elements related in neither direction.
Intersecting family	A collection of sets in which every two members have nonempty intersection.
Least element	An element ℓ satisfying $\ell \preceq x$ for every element x.
Linear order	Another name for a total order.
Maximal element	An element with no distinct element strictly above it.
Minimal element	An element with no distinct element strictly below it.
Partial order	A reflexive, antisymmetric, transitive relation.
Poset	A set together with a specified partial order.
Total order	A partial order in which every pair is comparable.

8 Likely exam questions

1. Verify a partial order

On $\mathbb{Z}^+$, define $a \preceq b$ when b/a is an integer. Prove this is a partial order.

Model answer

The condition is exactly $a \mid b$. Reflexivity follows from $a = a \cdot 1$. If $b = ak$ and $a = b\ell$ with positive integers k, ℓ, then $k\ell = 1$, so $a = b$. If $b = ak$ and $c = b\ell$, then $c = a(k\ell)$. Thus the relation is reflexive, antisymmetric, and transitive.

2. Find a precise counterexample

Why is divisibility not a partial order on all integers?

Model answer

Antisymmetry fails: $2 \mid -2$ and $-2 \mid 2$, but $2 \neq -2$. Reflexivity and transitivity still hold, but one failed property is decisive.

3. Construct a Hasse diagram

Find the covers for $S = \{1, 2, 4, 8, 16\}$ under divisibility.

Model answer

Every element divides the ones to its right, but only adjacent powers of two are covers:

$$(1, 2), (2, 4), (4, 8), (8, 16).$$

The Hasse diagram is a five-element chain. Edges such as $(1, 8)$ are omitted because intermediate elements exist.

4. Read a relation from a Hasse diagram

If a Hasse diagram has cover edges $a - b$, $a - c$, $b - d$, and $c - d$, list all ordered pairs in the partial order, reading upward.

Model answer

Include the four cover pairs $(a, b), (a, c), (b, d), (c, d)$, the transitive pair (a, d), and all four loops. Thus the relation has nine pairs total.

5. Separate maximal from greatest

In $S = \{2, 3, 6, 10\}$ under divisibility, find the minimal, maximal, least, and greatest elements.

Model answer

The minimal elements are 2 and 3. The maximal elements are 6 and 10. There is no least element because 2 and 3 are incomparable, and no greatest element because 6 and 10 are incomparable.

6. Prove uniqueness

Prove that a least element of a poset is unique.

Model answer

If ℓ_1 and ℓ_2 are both least, then $\ell_1 \preceq \ell_2$ because ℓ_1 is least, and $\ell_2 \preceq \ell_1$ because ℓ_2 is least. Antisymmetry yields $\ell_1 = \ell_2$.

7. Apply complement pairing

How many subsets can an intersecting family of subsets of a five-element set contain at most? Give a family attaining the bound.

Model answer

The $2^5 = 32$ subsets form 16 complement pairs, and at most one member of each pair can be selected. Thus the maximum is 16. All subsets containing a fixed element attain the bound.

9 Connections

- **From Lecture 15:** equivalence relations and partial orders both use reflexivity and transitivity. Symmetry produces classes; antisymmetry produces directed order.
- **To Lecture 17:** both relations and functions are subsets of a Cartesian product with extra rules. A partial order constrains which comparisons may coexist; a function constrains outputs per input.
- **To graph theory:** Hasse diagrams are directed acyclic structures with transitive edges removed. Upward reachability restores the order.
- **To scheduling:** prerequisites form a poset; incomparable tasks may proceed independently, while a chain gives a forced sequence.
- **To counting:** the complement-pair argument combines power sets, disjoint pairing, and the product rule.